

MUNDOSTREAMER

DJI FPV Video Receiver & Ground Station

OFFICIAL USER MANUAL & FIELD OPERATIONS GUIDE

Target Hardware	Radxa ROCK 3A (Rockchip RK3568 64-bit Quad-Core ARM)
Kernel & OS	Debian 12 Bookworm (Linux 6.1 Direct KMS Plane Engine)
Goggles Compatibility	DJI Goggles V1, V2, Goggles 2, Goggles Integra, Goggles 3
Display Output	HDMI 1080p @ 60 FPS (Hardware Direct Zero-Tear Plane)
Wireless Hotspot	Wi-Fi via USB Dongle (Radxa 3A lacks onboard Wi-Fi). Recommended: Ralink RT5370 / MT7601U
DVR Recording Format	Full HD 1080p H.265 (HEVC) MP4 + 3.5mm Onboard Mic Audio
Document Version	v2.4.0 Production Release — September 2026

[HIGHLIGHTS] WELCOME TO MUNDOSTREAMER

KEY HIGHLIGHTS & ARCHITECTURE ADVANTAGES:

- Zero-Latency HDMI Hardware Decoding: Video feed is displayed directly on monitors via Kernel Mode Setting (KMS) plane, bypassing desktop compositors.
- Full Visual Web UI & Screenshots: Complete step-by-step guidance illustrated with full screenshots across all 10 management tabs.
- Spectator WebRTC Live View: Spectators can watch instant low-latency live streams (<200ms) on smartphones without installing any apps.
- Safe Instant HDMI Replay: Replay recorded flight clips directly to pit monitors without disconnecting the DJI goggles USB communication.
- Onboard 3.5mm Mic Input: Record commentator/pilot audio directly mixed into DVR video files and network streams.
- Multi-Protocol Broadcasting: Native NDI auto-discovery, RTSP server, and UDP multicast for OBS Studio, vMix, and event production switchers.

Retain this manual for operational reference, maintenance protocols, and official technical support inquiries.

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CHAPTER 1: OVERVIEW OF MUNDOSTREAMER GROUND STATION

MundoStreamer is a standalone, enterprise-grade FPV Video Ground Station engineered specifically for drone racing pilots, freestyle enthusiasts, aerial cinematographers, and competitive FPV tournament organizers. The system bridges high-speed digital feeds from the DJI FPV goggles ecosystem directly to pit monitors (HDMI) and wirelessly broadcasts to spectator smartphones and production laptops simultaneously.

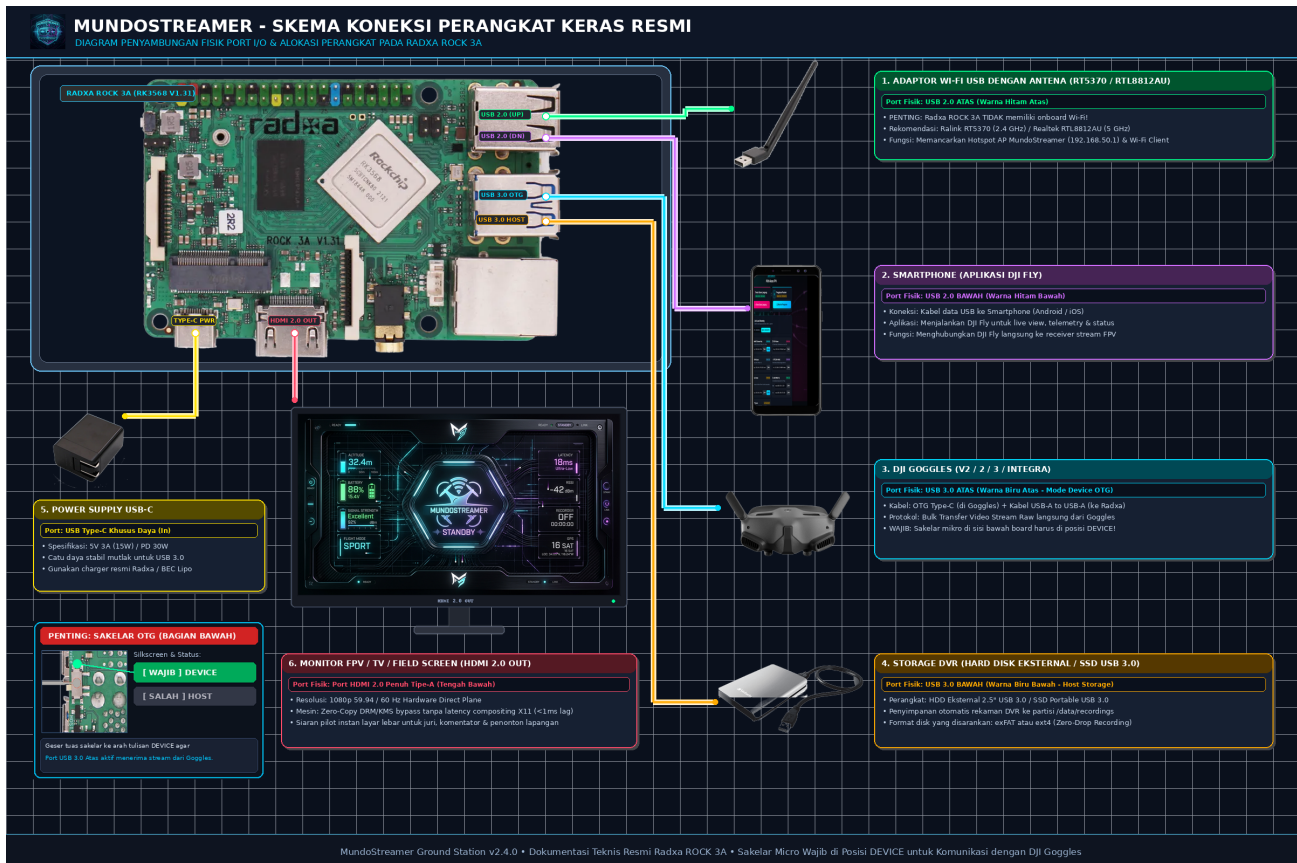
Why MundoStreamer Stands Out

Unlike generic Android-based screen mirroring or resource-heavy PC software converters that suffer from dropped frames and high latency, MundoStreamer runs on an embedded Linux architecture (Rockchip RK3568 SoC) with hardware-accelerated VPU decoding and direct Kernel Mode Setting (KMS) display planes. This architecture delivers decisive advantages:

- **Ultra-Low Latency:** Raw video packets from the goggles USB stream are piped straight to hardware VPU decoders with zero buffering overhead.
- **Silky 1080p60 Tear-Free Output:** HDMI display syncs directly to the display vertical refresh rate at 60 Hz, guaranteeing fluid high-speed motion.
- **Dual Simultaneous Feeds:** Pit spectators enjoy zero-lag HDMI while dozens of mobile spectators watch via Wi-Fi WebRTC/RTSP.
- **Intelligent Flight DVR:** Automatic arming/flight detection captures pure Full HD H.265 recordings with maximum storage efficiency.
- **Spectator Security Isolation:** Spectator mobile interfaces are strictly restricted to viewing, preventing accidental changes to pilot settings.

CHAPTER 2: PHYSICAL PORT LAYOUT & WIRING DIAGRAM

To ensure optimal performance and reliable USB gadget enumeration, follow the official hardware wiring architecture on the Radxa ROCK 3A board. The high-resolution diagram below illustrates the standard field ground station setup:



[CONNECTION GUIDE] PRIMARY CABLE ROUTING SUMMARY

Field Connection Quick Reference:

- Blue USB 3.0 UPPER (DJI Goggles): Attach USB-C OTG cable to Goggles, then connect via USB-A to USB-A cable to board. Micro-switch on PCB underside MUST BE IN DEVICE POSITION.
- Blue USB 3.0 LOWER (DVR Storage): Mounts external 2.5" HDD or USB 3.0 SSD for automatic flight recordings.
- Black USB 2.0 UPPER (Wi-Fi Dongle): Required to broadcast MundoStreamer Hotspot AP (IP: 192.168.50.1).
- Black USB 2.0 LOWER (Smartphone): Connect smartphone via USB data cable to run official DJI Fly app.
- High-Resolution Standalone Schema: Download printable 2-page poster at:

http://192.168.8.101:9876/downloads/Skema_Koneksi_Radxa3A.pdf

CHAPTER 2 (CONT'D): PORT SPECIFICATIONS & RULES

The following matrix outlines the bus interface specifications, kernel drivers, and mandatory usage guidelines for each physical port on the Radxa ROCK 3A:

PHYSICAL PORT	CONTROLLER TYPE	DEDICATED FUNCTION & GUIDANCE
Blue USB 3.0 UPPER (Top Blue Port)	DWC3 OTG Device	DEDICATED DJI GOGGLES PORT. Connection: Attach USB-C OTG cable to Goggles, then connect via USB-A to USB-A cable to this port. Operates in zero-latency Device OTG mode.
Blue USB 3.0 LOWER (Bottom Blue Port)	DWC3 xHCI Host 5 Gbps	DVR STORAGE MEDIA. Connect External 2.5" USB 3.0 Hard Drives or portable SSDs to store automatic Full HD flight recordings (path: /data/recordings).
Black USB 2.0 UPPER (Top Black Port)	Direct SoC USB Host	WI-FI ADAPTER PORT (ANTENNA). Houses external USB Wi-Fi dongle with high-gain antenna. Note: Radxa ROCK 3A has no onboard Wi-Fi. Recommended: Ralink RT5370 (external antenna) or RTL8812AU.
Black USB 2.0 LOWER (Bottom Black Port)	Terminus Hub USB 2.0	DJI FLY SMARTPHONE LINK. Connect smartphone running DJI Fly app via USB data cable to display telemetry, drone status, and live view.
HDMI Standard Out	HDMI 2.0 Direct Plane	MONITOR / TV OUTPUT. Connects to pit monitors, field TVs, projectors, or HDMI input goggles at 1080p60. Zero-lag hardware plane rendering (<1ms latency).
3.5mm TRRS Audio Jack	Rockchip RK809 Codec	ANALOG MIC & HEADPHONE. Supports analog clip-on microphones (CTIA standard) for live commentary and stereo headphone audio monitoring.
USB Type-C Power In	5V / 3A DC Input	MAIN DC POWER PORT. Requires a stable 5.0V / 3.0A power adapter or high-capacity power bank (15W minimum). Underpowered supplies cause USB 3.0 bus instability.

[CRITICAL WARNING] HARDWARE SWITCH MUST REMAIN IN DEVICE

CRITICAL HARDWARE WARNING: PCB UNDERSIDE SLIDE SWITCH

On the underside of the Radxa ROCK 3A PCB board, there is a small physical slide switch.

- This switch MUST ALWAYS REMAIN IN THE 'DEVICE' POSITION.
- If accidentally toggled to 'HOST', the Blue USB 3.0 UPPER port cannot receive stream from DJI Goggles.
- Never move this physical switch to Host!

CHAPTER 3: QUICK START GUIDE (POWER & SETUP)

Follow these 6 simple steps to power up and begin receiving live FPV video in under 2 minutes:

Step 1	Connect HDMI Display Cable Attach a high-speed HDMI cable from the MundoStreamer HDMI port to your monitor, television, or field display.
Step 2	Connect DJI Goggles USB Cable Use a premium high-speed USB-C to USB-A data cable. Connect the USB-C end to your DJI Goggles and the USB-A end to the Blue USB 3.0 UPPER port on MundoStreamer.
Step 3	Insert USB Storage (Optional) If you wish to record DVR video directly to a flash drive, insert a FAT32/exFAT drive into the Blue USB 3.0 LOWER port.
Step 4	Apply DC Power (Power On) Plug a 5V / 3A DC power source into the USB Type-C Power port. The onboard power LEDs will illuminate.
Step 5	Wait for HDMI Standby Video (~25s) The embedded Linux system performs a rapid boot. Within ~25 seconds, the monitor will display the official animated standby video: MundoStreamer - FPV Receiver.
Step 6	Power On DJI Goggles & Drone Turn on your DJI Goggles and quadcopter. As soon as the video RF link establishes, MundoStreamer instantly transitions from standby to Live 1080p60 FPV Video Output with zero lag!

[PRO TIP] CONNECTION RELIABILITY

CONNECTION & CABLE INTEGRITY TIPS:

- Always verify that the USB cable from DJI Goggles is a certified high-speed data cable (USB 2.0/3.0 High Speed), not a charge-only wire lacking data lines.
- If standby video remains after powering on the drone, open the Web UI and tap [Re-sync USB] in the top right header, or re-seat the goggles USB cable.

HDMI Screen Status Diagnostics

Your HDMI monitor gives immediate visual feedback on the system status:

- Looping Standby Animation: MundoStreamer is active and awaiting video data from DJI Goggles / drone.
- Smooth Live Video: Drone link established; video renders on direct hardware plane at 60 FPS.
- Black Screen / No Signal: Inspect HDMI cable seating or ensure the 5V 3A power adapter provides sufficient current.

CHAPTER 4: SMARTPHONE CONNECTION & WEB PORTAL

MundoStreamer features an integrated wireless Access Point and an intuitive Captive Portal system. Pilots and spectators can connect smartphones (Android, iPhone, iPad) or laptops wirelessly with zero app installs.

1. Connecting to the Wi-Fi Hotspot

- Open the Wi-Fi settings menu on your smartphone or tablet.
- Locate and select the SSID: MundoStreamer.
- Connect to the network (The default hotspot is Open / Password-Free).
- Upon connecting, a Captive Portal notification will automatically pop up on your device screen.
- If the prompt does not appear, open any web browser (Chrome / Safari) and navigate to: <http://192.168.50.1>

Captive Portal (Landing Page)

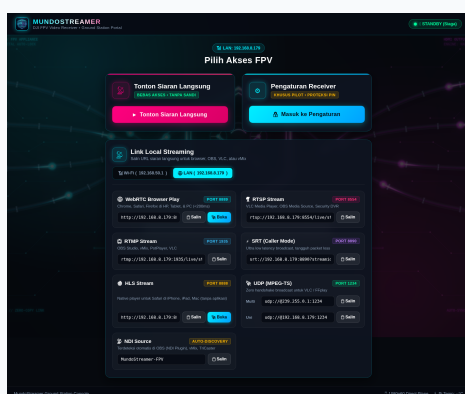


Figure 4.1: Spectator Portal & Pilot Entry

The default entrance page. Spectators can tap 'Spectator Live Stream' to watch with zero credentials.

Pilot PIN Authentication

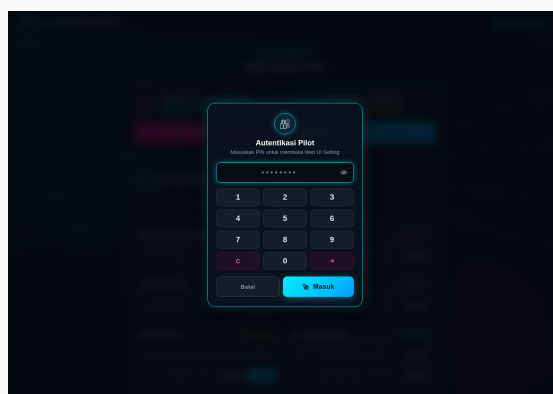


Figure 4.2: Pilot Settings PIN Keypad (11118888)

Security lock preventing unauthorized spectators from altering receiver settings. Enter default PIN 11118888 to enter.

[SECURITY PROTOCOL] PILOT PIN PROTECTION

PILOT SETTINGS PIN SECURITY (DEFAULT: 11118888)

To protect critical receiver configurations when dozens of spectators are on the Wi-Fi network, "Receiver Settings" is PIN-gated.

- Tap digits 11118888 on the virtual keypad or enter them using your device keyboard.
- You may customize this PIN anytime via Tab 10: System inside the management dashboard.

CHAPTER 5: WEB UI MENU & CONFIGURATION GUIDE

The MundoStreamer management dashboard follows international broadcast engineering standards in Strict English. Below are the core controls and recommended setups for Tab 1 and Tab 2:

Tab 1: Receiver Mode

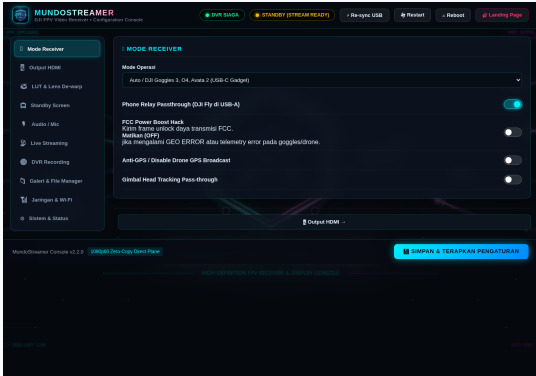


Figure 5.1: DJI Goggles Mode Configuration

Functions & Best Practices:

- **Auto-Detect Mode:** Automatically detects connected goggles (V1/V2/G2/Integra/G3).
- **Re-sync USB Button:** Cycles USB gadget drivers cleanly without system reboots.
- **Real-time Telemetry:** Shows live link status, frame counters, and FPS metrics.

Tab 2: HDMI Output

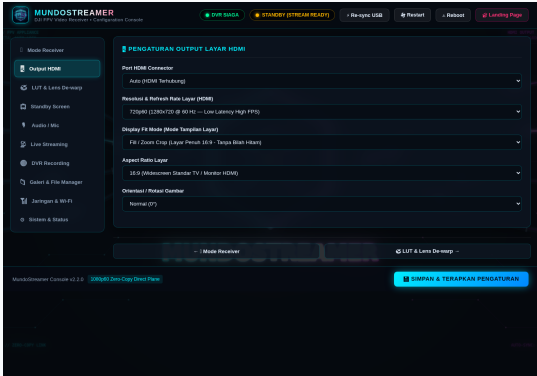


Figure 5.2: HDMI Display & Resolution Setup

Functions & Best Practices:

- **1080p @ 60 Hz:** Optimal crisp display preset for large pit monitors and event TVs.
- **720p @ 60 Hz:** Low-resolution option for legacy screens.
- **Aspect Ratio (16:9 / 4:3):** Select 4:3 for pilots accustomed to classic camera sensors.
- **HDMI Audio Passthrough:** Route live pilot audio directly to television speakers.

[NAVIGATION] HEADER CONTROLS

HEADER QUICK ACTION BUTTONS:

- **[Re-sync USB]:** Cycles USB gadget drivers without interrupting system services.
- **[Restart]:** Restarts video pipeline processes (dji-bridge and KMS renderer) if video hiccups.
- **[Reboot]:** Gracefully restarts the entire Radxa ROCK 3A host.
- **[Landing Page]:** Exits settings and returns to the public spectator entrance portal.

CHAPTER 5: WEB UI MENU GUIDE (CONTINUED)

Configuration guide for Tab 3 (LUT & Lens De-warp) and Tab 4 (Standby Screen & Branding):

Tab 3: LUT & Lens De-warp

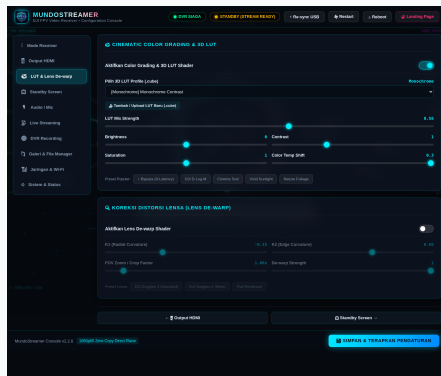


Figure 5.3: Lens Correction & Color Grading

Functions & Best Practices:

- **Lens De-warp (O3 / Action):** Straightens curved fisheye distortion from DJI O3 lenses for realistic horizon lines.
- **Color Profiles:** Choose Standard or Vivid enhancement curves.
- **Brightness & Contrast Sliders:** Adjust output dynamics for high-ambient sunlight pit monitors.

Tab 4: Standby Screen

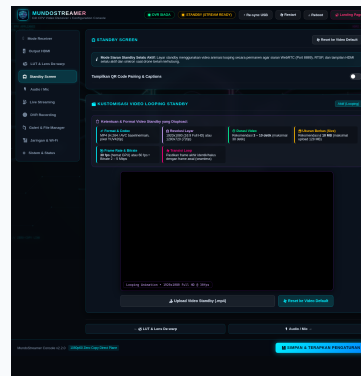


Figure 5.4: Standby Screen & Custom Branding

Functions & Best Practices:

- **Default Standby Animation:** MundoStreamer official looping brand animation.
- **Custom Video Upload:** Upload your own MP4 video for tournament sponsors, team intro reels, or racing promos.
- **Custom Wallpaper:** Display static team logos or high-res graphics.

[EVENT BRANDING] STANDBY VIDEO GUIDELINES

CUSTOM STANDBY VIDEO SPECIFICATIONS:

For optimal visual presentation on HDMI displays, use an MP4 file formatted at 1920x1080 @ 60 FPS encoded with H.264 or H.265. The video automatically loops seamlessly whenever no live drone feed is active.

CHAPTER 5: WEB UI MENU GUIDE (CONTINUED)

Configuration guide for Tab 5 (Audio / Mic Input) and Tab 6 (Live Network Streaming):

Tab 5: Audio / Mic

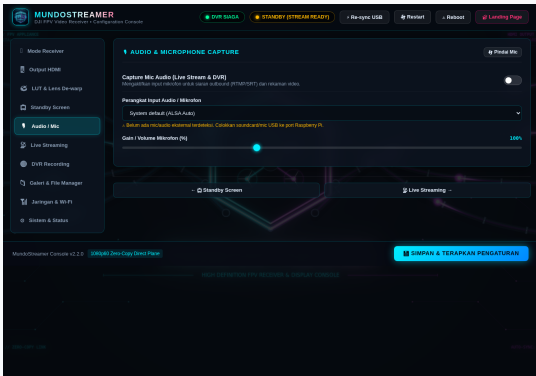


Figure 5.5: Microphone & Audio Mixer Settings

Functions & Best Practices:

- **Enable 3.5mm Microphone:** Activates analog input from the onboard TRRS jack (RK809 codec).
- **Mic Gain (+0dB to +20dB):** Amplifies clip-on condenser mics without outboard hardware.
- **Pilot Voice Muxing:** Seamlessly mixes voice commentary into DVR MP4s and live network streams.

Tab 6: Live Streaming

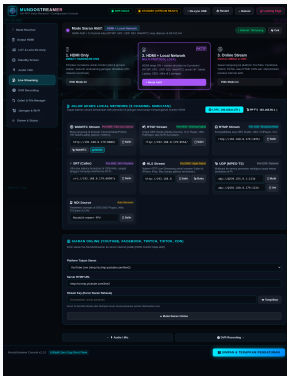


Figure 5.6: Network Streaming Protocols

Functions & Best Practices:

- **WebRTC MediaMTX (Port 8889):** Sub-200ms ultra-low latency playback in mobile browsers.
- **RTSP Server (Port 8554):** Industry-standard stream for OBS Studio and VLC.
- **NDI Broadcaster:** Zero-configuration network stream named MundoStreamer-FPV.
- **UDP Multicast / Unicast:** Integration with IPTV networks and switcher cards.

[STREAMING BEST PRACTICES] PROTOCOL RECOMMENDATIONS

STREAMING PROTOCOL SELECTION GUIDE:

- For Spectators on Mobile in Pit Areas: Use WebRTC — instant playback in browsers without downloading any third-party app.
- For Production Laptops running OBS Studio / vMix: Use NDI for zero-configuration discovery and visually lossless 60 FPS delivery.

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CHAPTER 5: WEB UI MENU GUIDE (CONTINUED)

Configuration guide for Tab 7 (DVR Recording) and Tab 8 (Gallery & Storage Management):

Tab 7: DVR Recording

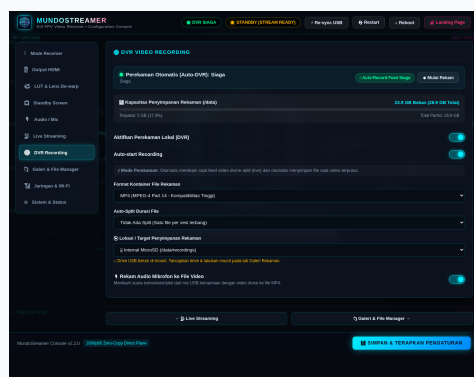


Figure 5.7: H.265 DVR Recording Engine

Functions & Best Practices:

- **Enable Auto-DVR:** Automatically begins recording when quad arms/takes off and finishes on disarm.
- **Manual Start/Stop Buttons:** Manual override controls with live recording duration counters.
- **Audio Mode:** Toggle between Pure Video (H.265) or Video + Pilot Mic Audio.
- **Storage Target:** Store to internal eMMC or external high-speed USB 3.0 drives.

Tab 8: Gallery & Files

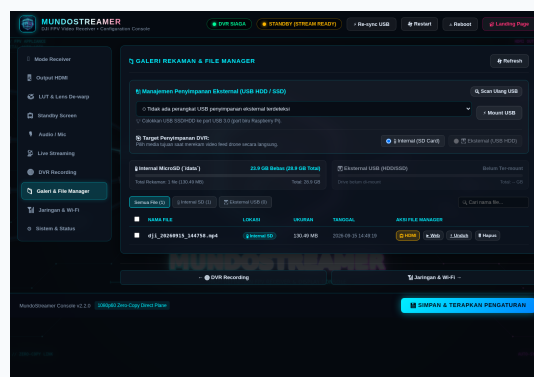


Figure 5.8: DVR File Gallery & HDMI Playback

Functions & Best Practices:

- **Play on HDMI Button:** Plays recorded flights directly to pit monitors without disconnecting goggles.
- **Download MP4 Button:** Download recorded files directly to smartphones or laptops.
- **Delete Button:** Remove older video files to free up disk space.

[EXCLUSIVE FEATURE] INSTANT HDMI REPLAY

ZERO-DISCONNECT INSTANT HDMI REPLAY:

In Tab 8, tap the blue [Play on HDMI] button beside any flight video to replay it immediately on the HDMI monitor. If a pilot powers on another quadcopter, the system automatically terminates replay and resumes live flight view instantly!

CHAPTER 5: WEB UI MENU GUIDE (CONTINUED)

Configuration guide for Tab 9 (Network & Wi-Fi) and Tab 10 (System Diagnostics & Power):

Tab 9: Network & Wi-Fi

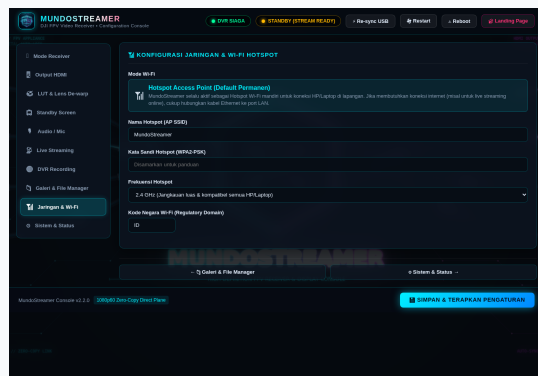


Figure 5.9: Wi-Fi Hotspot & Network Setup

Functions & Best Practices:

- **Hotspot SSID & Channel:** Adjust Wi-Fi name and RF channel to avoid crowded 2.4GHz bands at racing venues.
- **Wi-Fi Client Mode:** Connect MundoStreamer to venue Wi-Fi or smartphone tethering for cloud broadcasts.
- **IP Status Table:** Displays active IP addresses for Hotspot and wired Gigabit LAN.

Tab 10: System & Telemetry

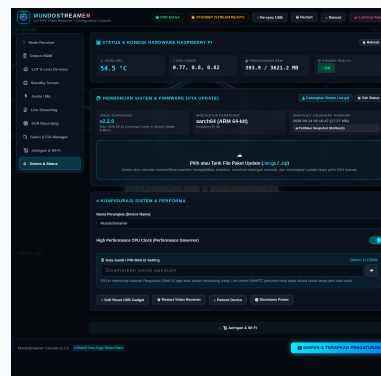


Figure 5.10: Diagnostics, Licensing & Power

Functions & Best Practices:

- **Hardware Telemetry:** Real-time monitoring of RK3568 SoC temperatures and RAM usage.
- **Download User Manual:** Download this official PDF manual directly to any device.
- **License Manager:** View permanent hardware-bound Machine ID and PRO status.
- **Power Off System Button:** Safe shutdown before disconnecting power cables.

[HARDWARE RECOMMENDATION] OFFICIAL USB WI-FI ADAPTER GUIDE

USB WI-FI ADAPTER GUIDE (RADXA ROCK 3A HAS NO ONBOARD WI-FI):

Unlike Raspberry Pi, the Radxa ROCK 3A single-board computer does not have built-in onboard Wi-Fi or Bluetooth. To broadcast the MundoStreamer Wi-Fi Hotspot and Web Portal, users must connect a compatible USB Wi-Fi dongle to the Black USB 2.0 UPPER port.

- Official Tested Dongle (Factory Standard & Highly Recommended):
 - Ralink / MediaTek RT5370 (USB ID: 148f:5370, Driver: rt2800usb). Features external 2.4GHz SMA stick antenna. Proven rock-solid stability for AP Hotspot mode (zero disconnects), broad outdoor range, and 100% Plug & Play out-of-the-box with Linux mainline kernel.
 - Other Supported Chipsets on Radxa ROCK 3A Kernel:
 1. MediaTek MT7601U (Driver: mt7601u) — Inexpensive 2.4GHz alternative (e.g. TP-Link TL-WN727N v5, Mercury MW150US, generic dongles).
 2. Realtek RTL8188EUS / RTL8188FTV (Driver: rtl8xxxu) — Standard 2.4GHz nano or antenna USB adapters.
 3. Realtek RTL8811AU / RTL8812AU — Dual-Band AC (2.4GHz & 5GHz) for high spectator streaming throughput.
 4. Radxa Wireless Module A8 (M.2 E-Key) — Internal PCIe Wi-Fi 6 module (RTL8852BE, driver rtlw89_8852be).
- Note: Always insert the USB Wi-Fi adapter into Black USB 2.0 UPPER before powering on so that interface wlan0 initializes automatically.

[OPERATIONAL PROTOCOL] SAFE SHUTDOWN PROCEDURE

SAFE SYSTEM SHUTDOWN PROTOCOL:

Never yank the DC power cable while the operating system is actively writing to eMMC storage or USB media!

- Always navigate to Tab 10: System and click the red Power Off System button.
- Wait 10-15 seconds until the HDMI display goes completely black and onboard LEDs cease blinking before removing power.

CHAPTER 6: DVR RECORDING & INSTANT HDMI REPLAY

MundoStreamer is equipped with a professional DVR recording engine capable of capturing pure 1080p60 H.265 video with hardware timestamping and fast-start MP4 container generation.

1. Automatic Flight DVR (Auto-Record)

When Enable Auto-DVR is active in Tab 7, the system monitors the DJI OSD telemetry data stream. The moment the drone is armed, recording triggers automatically with zero user intervention. When the drone lands and disarms, the file is cleanly finalized and indexed into the gallery.

2. Storage Media Selection

- External USB 3.0 Drive (Recommended): Insert flash drive into Blue USB 3.0 LOWER port. Video files are written directly to the drive, allowing instant transfer to editing workstations.
- Internal eMMC Memory: If no USB drive is plugged in, recordings are securely stored in the internal 16GB/32GB eMMC partition.

3. Revolutionary Feature: Zero-Disconnect Instant HDMI Replay

Beginning in v2.4.0, MundoStreamer introduces Native Standby Symlink Injection technology. You can replay recorded flight runs directly onto pit monitors without breaking the USB connection to DJI Goggles!

Step	Operational Replay Instructions
Step 1	Navigate to Tab 8: Gallery & Files on the Web UI.
Step 2	Locate the desired flight recording from the file list.
Step 3	Click the blue [Play on HDMI] button next to the filename.
Step 4	The video replays immediately on the HDMI monitor at silky 60 FPS. The Web UI displays an active banner: Playing on HDMI: [filename.mp4].
Step 5	To conclude video replay and return to standby animation, click [Stop HDMI Playback].

[FAIL-SAFE] LIVE FLIGHT PRIORITY

LIVE FLIGHT PRIORITY OVERRIDE:

You never need to worry about interrupting an upcoming heat if someone is watching a replay on the pit screen. MundoStreamer implements automatic live flight priority: The microsecond live RF packets from a powered drone are detected, HDMI replay halts instantly and the monitor transitions directly to live FPV video!

CHAPTER 7: STREAMING TO OBS, VMIX & WEBRTC

MundoStreamer is the ultimate FPV receiver solution for tournament broadcasting, live streaming, and content creators. It provides 3 concurrent video feeds to PCs, OBS Studio, vMix, and mobile devices over standard network infrastructure.



Figure 7.1: Mobile WebRTC Spectator Player Interface (Player.html) with <200ms Latency

Method 1: Using NDI Source (Recommended for Broadcast)

NDI (Network Device Interface) by NewTek delivers broadcast-grade, low-latency streaming with zero manual IP setup.

1. Ensure the production laptop connects to the MundoStreamer Wi-Fi or same Gigabit LAN switch.
2. Launch OBS Studio or vMix on your laptop.
3. In OBS Sources, click + and select NDI Source.
4. In the Source name dropdown, choose MundoStreamer-FPV.
5. Click OK. Crystal-clear 1080p60 video appears instantly without typing IP addresses!

Method 2: Using RTSP Media Source

1. Open OBS Studio -> Click + in Sources -> Select Media Source.
2. Uncheck Local File.
3. Enter URL: `rtsp://192.168.50.1:8554/live` (or LAN IP). Set network buffering to 0-50 ms.

Method 3: Using WebRTC Mobile Browser Player

Connect spectator phones to the MundoStreamer Wi-Fi and open:

`http://192.168.50.1:8889/live/stream/?autoplay=true`. The video stream plays immediately with ultra-low latency under 200 milliseconds.

Protocol	Latency	Compatibility	Primary Advantages
NDI Source	< 100 ms	OBS (NDI Plugin), vMix, TriCaster	Zero IP config, auto-discovery, peak 1080p60 fidelity.
RTSP Stream	< 200 ms	OBS Media Source, VLC, PotPlayer	Universal standard, rock-solid over long Ethernet cable runs.
WebRTC Player	< 150 ms	Google Chrome, Apple Safari, Firefox	Instant mobile playback with zero apps required.

CHAPTER 8: COMMENTATOR MIC & 3.5MM AUDIO

MundoStreamer features onboard analog audio processing powered by the Rockchip RK809 high-fidelity codec. This allows race commentators and pilots to record real-time audio commentary directly into DVR recordings and live feeds.

1. Hardware Microphone Setup

- Connect an analog lapel / clip-on microphone to the 3.5mm TRRS jack on the MundoStreamer board.
- The jack utilizes the standard CTIA (Cellular Telecommunications Industry Association) pinout (Tip: Left, Ring1: Right, Ring2: Ground, Sleeve: Mic).

2. Enabling Microphone in Web UI

1. Open the Web UI and enter Pilot PIN 11118888.
2. Navigate to Tab 5: Audio / Mic.
3. Switch the toggle Enable 3.5mm Microphone to ON.
4. Adjust the Mic Gain Slider to achieve clear voice projection without distortion (Recommended: +10 dB to +15 dB).
5. Go to Tab 7: DVR Recording -> Ensure Audio Mode is set to Video + Mic Audio.
6. Begin flight recording. Voice commentary is cleanly muxed into the MP4 file at pristine 48kHz AAC fidelity!

[AUDIO ENGINEERING] MIC BEST PRACTICES

AUDIO FIDELITY TIPS:

- If utilizing a standard 2-ring (TRS) computer microphone, use a TRS to CTIA TRRS adapter so the mic signal maps properly to the sleeve terminal.
- In Tab 5, you can also route the microphone audio out to pit speakers so spectators around the tent can hear live race announcements.

CHAPTER 9: MAINTENANCE & TROUBLESHOOTING

If you encounter unexpected behavior in the field, consult this quick reference troubleshooting matrix before contacting technical support:

Symptom	Potential Cause	Practical Solution
HDMI Screen Black / No Signal	- Loose HDMI cable. - Power source below 5V 3A. - Monitor set to wrong HDMI port.	- Check monitor input source. - Use a certified high-speed HDMI cable. - Verify power adapter supplies true 5.0V / 3.0A current.
HDMI Stuck on Standby Animation	- Goggles plugged into wrong port. - Cable is charge-only. - Underside switch moved to Host.	- Ensure goggles are in Blue USB 3.0 UPPER port. - Verify underside switch is set to OTG. - Tap [Re-sync USB] in Web UI.
MundoStreamer Wi-Fi Not Visible	- Wi-Fi dongle loose / missing. - Wrong USB port or unsupported chipset.	- Radxa 3A lacks onboard Wi-Fi, ensure USB dongle is connected. - Plug into Black USB 2.0 UPPER port. - Use recommended chipset (Ralink RT5370 / MT7601U).
USB Drive Not Detected for DVR	- File system not FAT32/exFAT. - Plugged into top USB port.	- Format flash drive to FAT32 or exFAT on PC. - Insert into Blue USB 3.0 LOWER port.
Forgotten Web UI Pilot PIN	PIN was changed from factory default (11118888).	Contact official MundoStreamer support via WhatsApp (+62 856-4722-9919) for PIN recovery instructions.

Thermal Management & Hardware Care

- Airflow & Cooling:** Ensure the cooling heatsink fan on the Radxa ROCK 3A board is free from dust and obstructions. Ideal operating CPU temperature is between 45°C - 65°C.
- Moisture Protection:** The unit is built for outdoor pit use but is not waterproof. Shield the receiver from rainfall or heavy morning condensation.
- Protective Enclosure:** Store the unit in a padded travel case during transit to safeguard antenna connectors and the underside OTG switch.

CHAPTER 10: OFFICIAL LICENSING & SUPPORT

MundoStreamer is safeguarded by an enterprise hardware-bound licensing architecture (Hardware Machine ID Lock V2) permanently keyed to the cryptographic silicon signature of your Rockchip RK3568 processor.

1. Identifying Your Machine ID

Each MundoStreamer hardware station carries a unique cryptographic identifier formatted as:

MUNDO-R3A-XXXX-XXXX-XXXX-XXXX

You can inspect this code on the landing portal or under Tab 10: System. The license is permanent and immune to micro-SD card replacements, Wi-Fi dongle changes, or operating system updates.

2. Warranty Coverage & Updates

- Hardware Warranty: 1-year limited manufacturer warranty covering factory defects and component reliability.
- Lifetime Software Updates: Licensed owners receive continuous OTA firmware updates and feature additions.
- Configuration Backup: Always export your system configuration backup via Tab 10: System before applying system updates.

[OFFICIAL SUPPORT] GET IN TOUCH

OFFICIAL TECHNICAL SUPPORT & CUSTOMER SERVICE:

For activation assistance, event broadcast consultation, or technical inquiries:

- WhatsApp Official Support: +62 856-4722-9919 (Direct Priority Support)
- Telegram Pilot Community: @mundostreamer_fpv
- Local Firmware Portal: <http://192.168.50.1/manual> (Direct Device Link)
- Technical Support Email: support@mundostreamer.id

When contacting technical support, please attach a photograph of your unit's Machine ID for instant ownership verification.

MundoStreamer Engineering Team

Dedicated to the Global FPV Pilot Community & Broadcast Professionals